

Complete Design Report

The full concept and preliminary design proposal.

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VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository	github.com/wasim437/AI_SAFA
Live portal	wasim437.github.io/AI_SAFA/
Produced by	run_analysis.py · src/plan.py · src/costing.py · models/headline_metrics.json
Reproduce	<code>python run_analysis.py</code> · <code>python -m tests.test_pipeline</code>

How this document was produced

This submission is the output of a ten-phase process in which each phase is checked against the one before it. Nothing downstream is hand-drawn or hand-typed: change an input, re-run, and every chart, drawing and figure moves with it — or a test fails loudly. The phases behind this particular document are marked.

Phase 1 — Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

Phase 2 — Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

Phase 3 — Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

Phase 4 — Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

Phase 5 — Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

Phase 6 — Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

Phase 7 — Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

Phase 8 — User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

Phase 9 — AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

Phase 10 — Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

THE CODE AND DATA BEHIND THIS DOCUMENT

[run_analysis.py](#)

[src/plan.py](#)

[src/costing.py](#)

[models/headline_metrics.json](#)

REPRODUCE IT

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 38 checks
```

Complete Design Report

Falaj Al Safa — concept and preliminary design for Al Safa 2 Park

A 15,000 m² neighbourhood park in Dubai, redesigned around one question: how do you make an outdoor space usable in a city that is too hot to stand in? The answer proposed here raises comfortable daylight hours from 44.5% to 64.6%, within a capital cost of AED 27.0 M against the AED 35 M budget.

Upload slot 10 — Complete Design Report

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai

Mohamed Wasim · Individual Applicant

Issued 03 August 2026

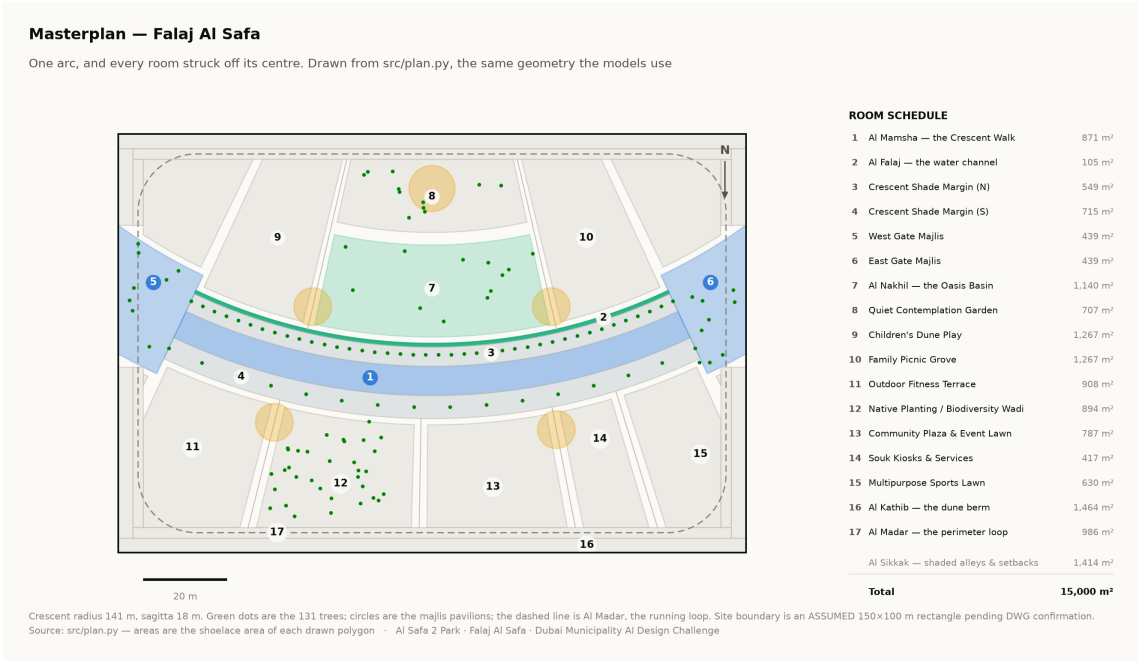
Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. The scheme

One arc. A crescent of shade 141 m in radius sweeps across the site, bowing convex south so that its hollow opens to the north. A 0.9 m water channel runs beneath its northern drip line. Every room in the park is struck off the same centre, so no room is a rectangle and every room faces the crescent square-on.

Element	What it is	Description
Al Hilal	the Crescent Canopy	18 m gridshell at 4.5 m over a 7 m walk, with a 3 m southern louvre
Al Falaj	the water channel	0.9 m wide, on the canopy's drip line so it is shaded all day and evaporates less
Al Nakhil	the Oasis Basin	a sunken palm court held in the crescent's concave side
Al Sikkak	the alleys	radial — each one a radius of the same arc
Al Madar	the perimeter loop	a tree-shaded running and walking circuit
Al Kathib	the dune berm	planted earth against the roads — noise, glare and heat

The six named elements. All are generated from a single arc definition in src/plan.py.



Masterplan. Technical drawing — to scale.

2. Performance

Measure	Today	As designed
Comfortable daylight hours	44.5%	64.6%
Peak heat index	56.8 °C	48.7 °C
Crescent Walk shaded	—	87.3%
Site-wide mean shade	—	34.1%
Trees	—	131

3. Room schedule

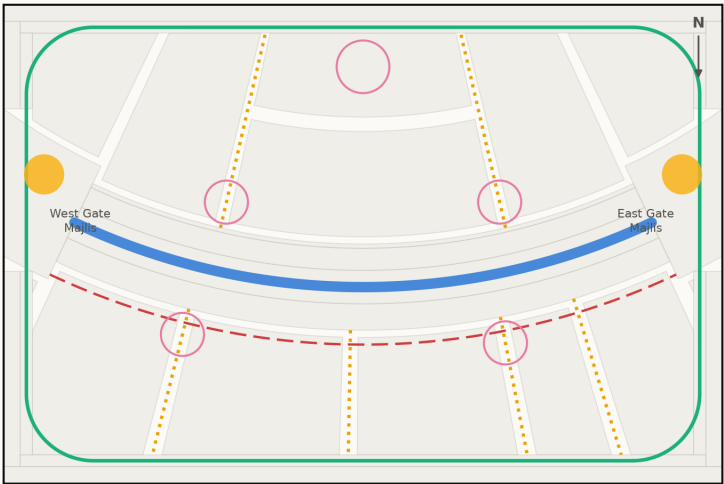
Room	Category	Area (m²)	% of site
Al Mamsha — the Crescent Walk	Circulation	871	5.81%
Al Falaj — the water channel	Water	105	0.70%
Crescent Shade Margin (N)	Green	549	3.66%
Crescent Shade Margin (S)	Green	715	4.77%
West Gate Majlis	Arrival	439	2.93%
East Gate Majlis	Arrival	439	2.93%
Al Nakhil — the Oasis Basin	Green	1,140	7.60%
Quiet Contemplation Garden	Passive	708	4.72%
Children's Dune Play	Active	1,267	8.45%
Family Picnic Grove	Passive	1,267	8.45%
Outdoor Fitness Terrace	Active	908	6.06%
Native Planting / Biodiversity Wadi	Green	894	5.96%
Community Plaza & Event Lawn	Social	788	5.25%
Souk Kiosks & Services	Commercial	417	2.78%
Multipurpose Sports Lawn	Active	630	4.20%
Al Kathib — the dune berm	Green_Buffer	1,464	9.76%
Al Madar — the perimeter loop	Circulation	986	6.57%
Al Sikkak — shaded alleys & setbacks	Circulation	1,414	9.43%

Total 15,000 m².

4. Circulation, accessibility and facilities

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



- Al Mamsha — primary route, shaded, step-free

Al Sikkak — shaded alleys to the rooms

Al Madar — 438 m running loop
- Service / emergency vehicle access

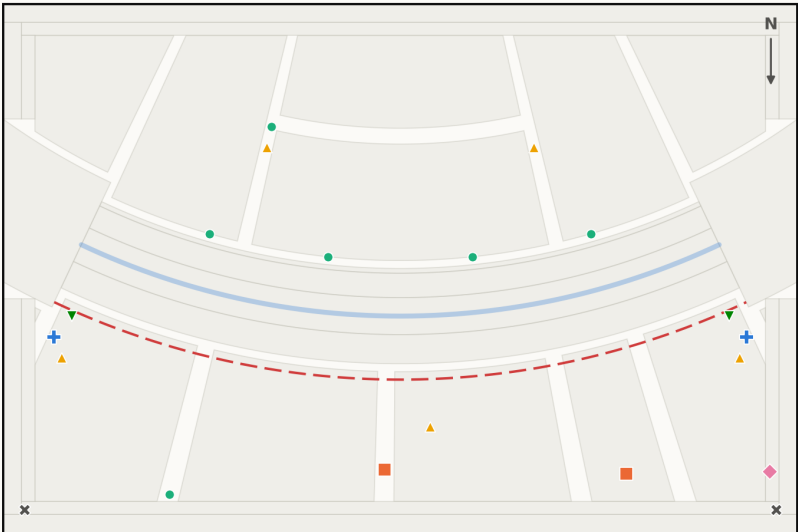
Majlis — shaded rest, never more than 33 m from the walk

Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped. Source: src/plan.py — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Circulation and accessibility. Technical drawing — to scale.

Commercial & service facilities

Where the park earns its running costs, and where it is serviced



- Service / emergency vehicle access

Commercial — souk kiosks and café (2)
- Drinking fountain (6)

Waste & recycling station (5)

Service to maintenance zone (1)
- Bicycle parking (2)

Drop-off / pick-up bay (2)

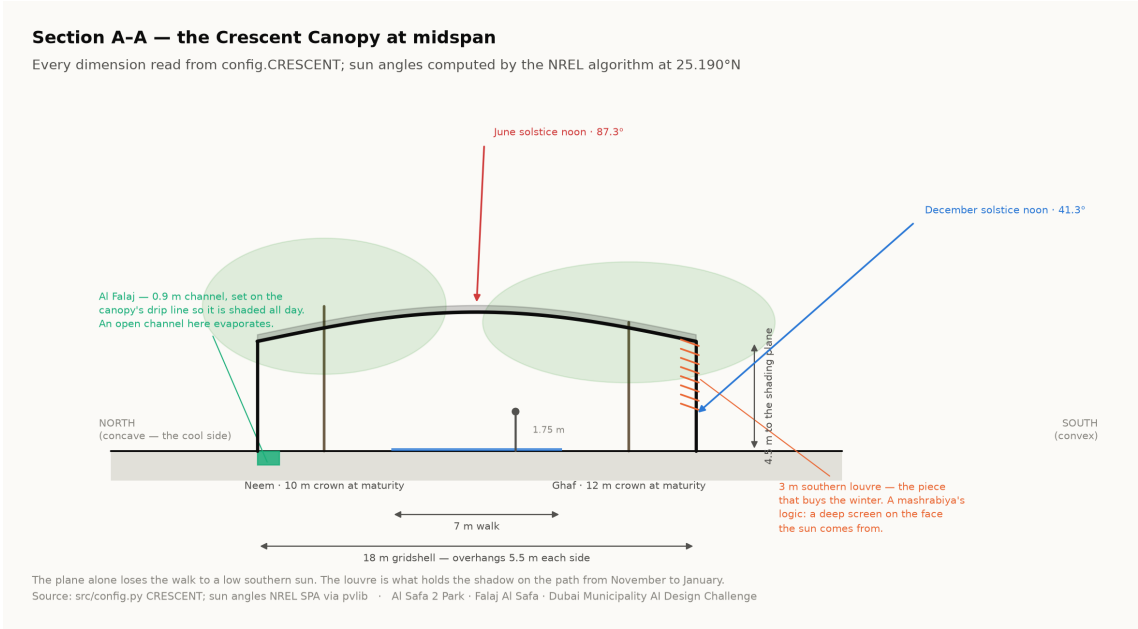
Public restrooms (2)

20 facilities. Commercial floor slabs are 100 m² each, 2.8% of the site. The souk face (417 m², 2.8% of the site) is the main revenue source, so that evening trade and programming reinforce each other — and away from the concave hollow, which is kept quiet. Every facility is reachable from the dashed service route without a vehicle entering the walk. Source: src/plan.py — facility positions are arc coordinates, so they move with the crescent · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Commercial and service facilities, with the service route. Technical drawing — to scale.

20 facilities are placed against the drawn geometry: 2 commercial, 2 restroom blocks, 6 drinking fountains, 5 waste and recycling points, 1 maintenance store, 2 bicycle parking areas and 2 drop-off bays. Each is reachable from the service route without a vehicle entering the walk.

5. The section that makes it work



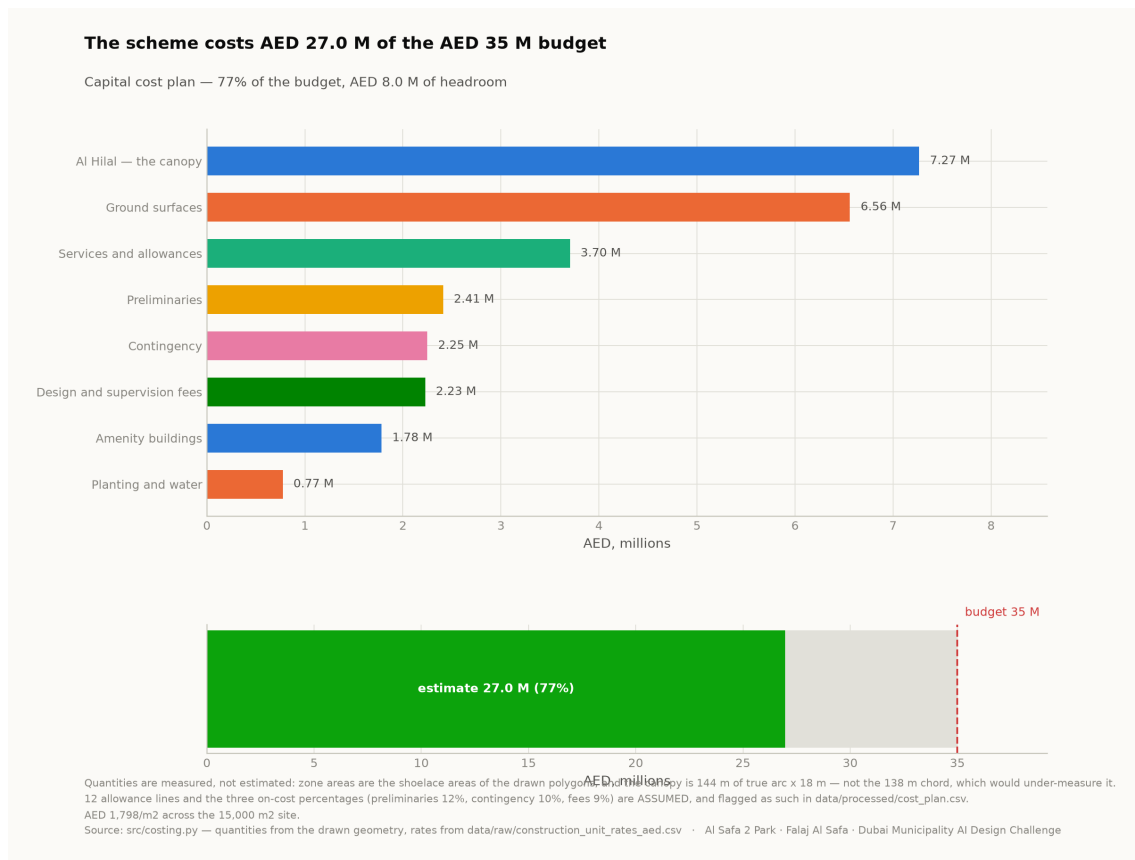
Section A–A: a 7 m walk under an 18 m gridshell at 4.5 m, with a 3 m southern louvre. Sun angles computed, not drawn. Technical drawing — to scale.

6. Feasibility — cost against the AED 35 M budget

The capital cost plan measures quantities from the drawn geometry rather than estimating them. Zone areas are shoelace areas; the canopy is priced on 144.2 m of *true arc* — not the 138 m chord, which would under-measure the structure by more than 6 m.

Element	AED millions
Al Hilal — the canopy	7.27
Ground surfaces	6.56
Services and allowances	3.70
Amenity buildings	1.78
Planting and water	0.77
Preliminaries	2.41
Contingency	2.25
Design and supervision fees	2.23
TOTAL	26.97
Budget	35.00
Headroom	8.03

Total AED 26,973,013, or 77.1% of the budget, at AED 1,798/m² across the site. Allowance lines and the three on-cost percentages are flagged ASSUMED in data/processed/cost_plan.csv.



Capital cost plan against the budget. Analysis output — computed from project data.

7. Honesty notes

- The **site boundary is assumed** — a 150 × 100 m rectangle pending confirmation against the issued CAD file. Every area figure depends on it.
- The **climate series is modelled** from 39 years of monthly normals, verified back to within 0.39 °C, and labelled as modelled everywhere it appears.
- **Visitor demand is a scenario**, not a prediction, and is deliberately excluded from the machine learning suite.
- **Diffuse radiation is excluded** from the shade model, so shaded comfort is stated conservatively.
- The **photovoltaic array runs a deficit**, covering about 13% of load — not a surplus.
- The **renders are illustrations**, not photographs of a built thing, and none is presented as analysis.

8. Corrections this submission makes to itself

An earlier version of this project claimed 99.2% annual shade. It did not survive a geometric check and was withdrawn; the re-solved section measures 87.3%. Three visuals that presented invented data as measurement were also withdrawn. Both corrections are stated here rather than quietly removed, because a submission that audits itself in public is more credible than one that never had to.

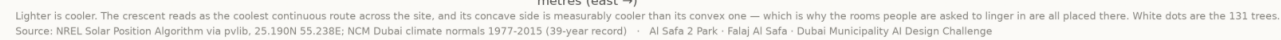
Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Presentation board.

Dubai Municipality AI Park Design Challenge | Board 2 of 2 — EVIDENCE

Predicted summer comfort across the site

July afternoon heat index per square metre — the shade strategy, tested



What actually determines whether a square metre is shaded

Permutation importance — the drop in accuracy when each feature is shuffled



Planting plan – 131 trees

Drawn at mature canopy radius. The avenue is structural; it carries the shoulder seasons the gridshell cannot



28 of 131 trees are UAE natives (Ghaf and Date Palm). Peak summer irrigation demand 11.8 m³/day. Ghaf takes the southern rank because it is the most drought-tolerant species in the schedule and that rank is the more exposed of the two.

Shade converts unusable daylight hours into usable ones

Share of the year's 4,402 daylight hours in each comfort band



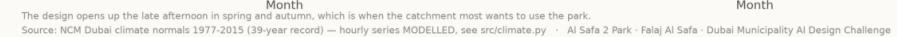
Annual shade coverage by zone type

Ray-traced ground-plane occlusion, 1 m grid, sampled across the year



When is the park comfortable?

Mean heat index by hour and month — exposed today, shaded as designed



Elevation — the Crescent Canopy
60 m of the 124 m run, to scale, 6 m structural bay

60 m of the 124 m run, to scale, 6 m structural bay



Comfortable daylight hours rise from 44.5% to 64.6% — 20.1 points of the year handed back. Peak heat index falls 56.8 °C → 48.7 °C.

Site-wide mean shade is 34.1%, and that is stated as a position rather than hidden: this scheme makes a few places genuinely excellent instead of the whole site marginally better.

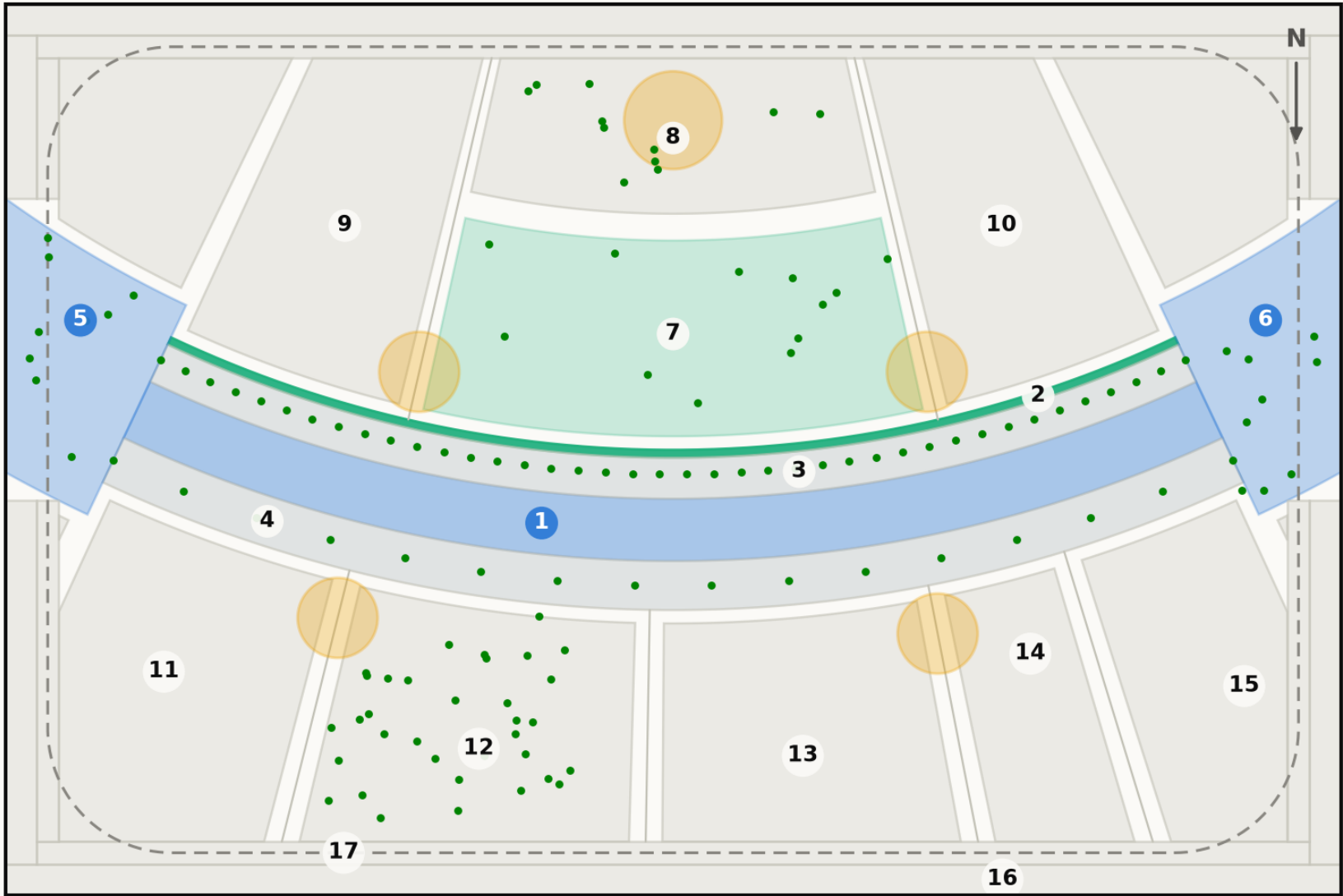
Shade surrogate R^2 0.994 · comfort classifier 97.5% with temperature and humidity withheld · 2 microclimate regimes, k selected by silhouette

Fig10 masterplan

Technical drawing — to scale. Geometry generated by src/plan.py; every area is the measured area of the drawn polygon.

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



20 m

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

ROOM SCHEDULE

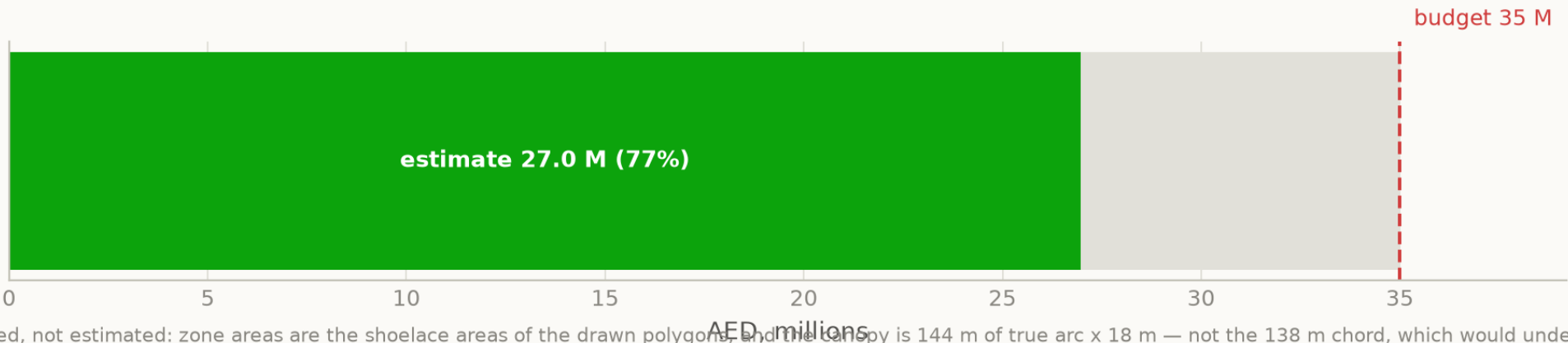
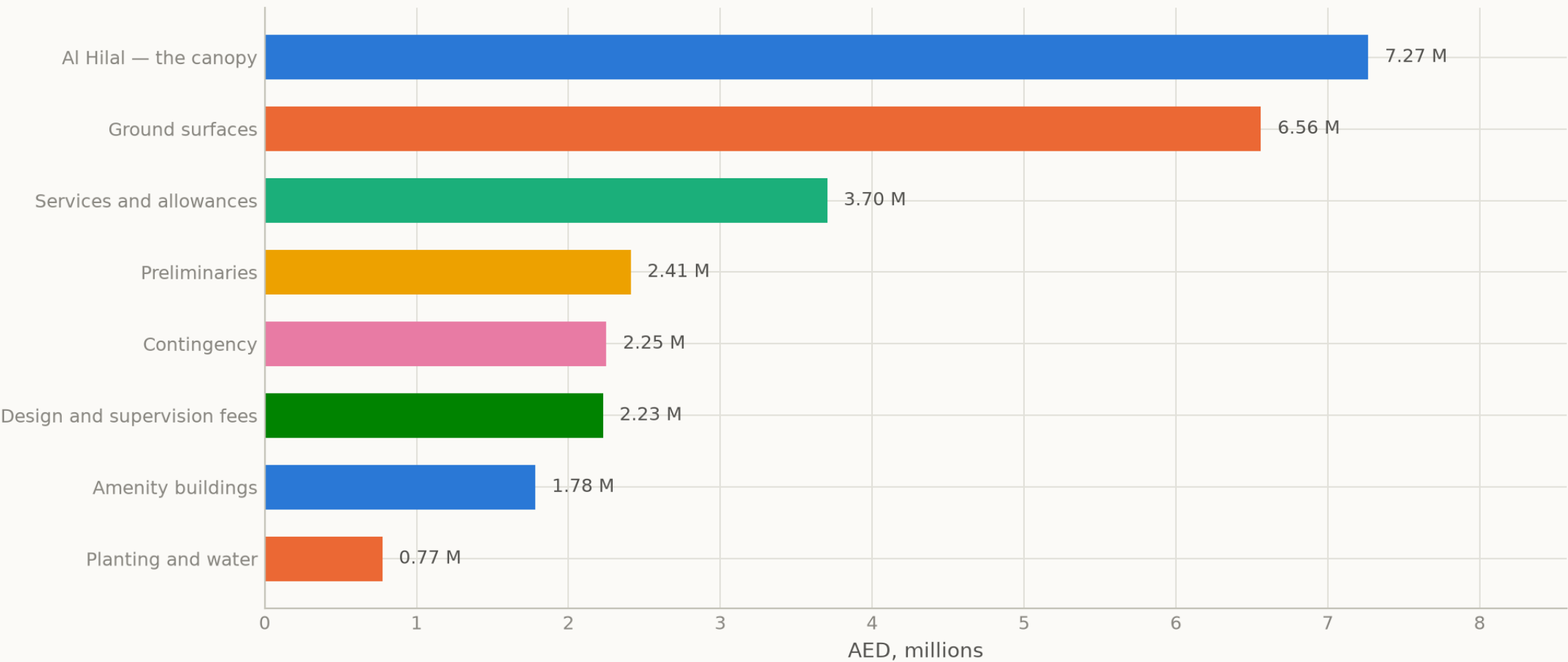
1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
Al Sikkak — shaded alleys & setbacks		1,414 m²
Total		15,000 m²

Fig11 cost plan

Analysis output — computed from project data.

The scheme costs AED 27.0 M of the AED 35 M budget

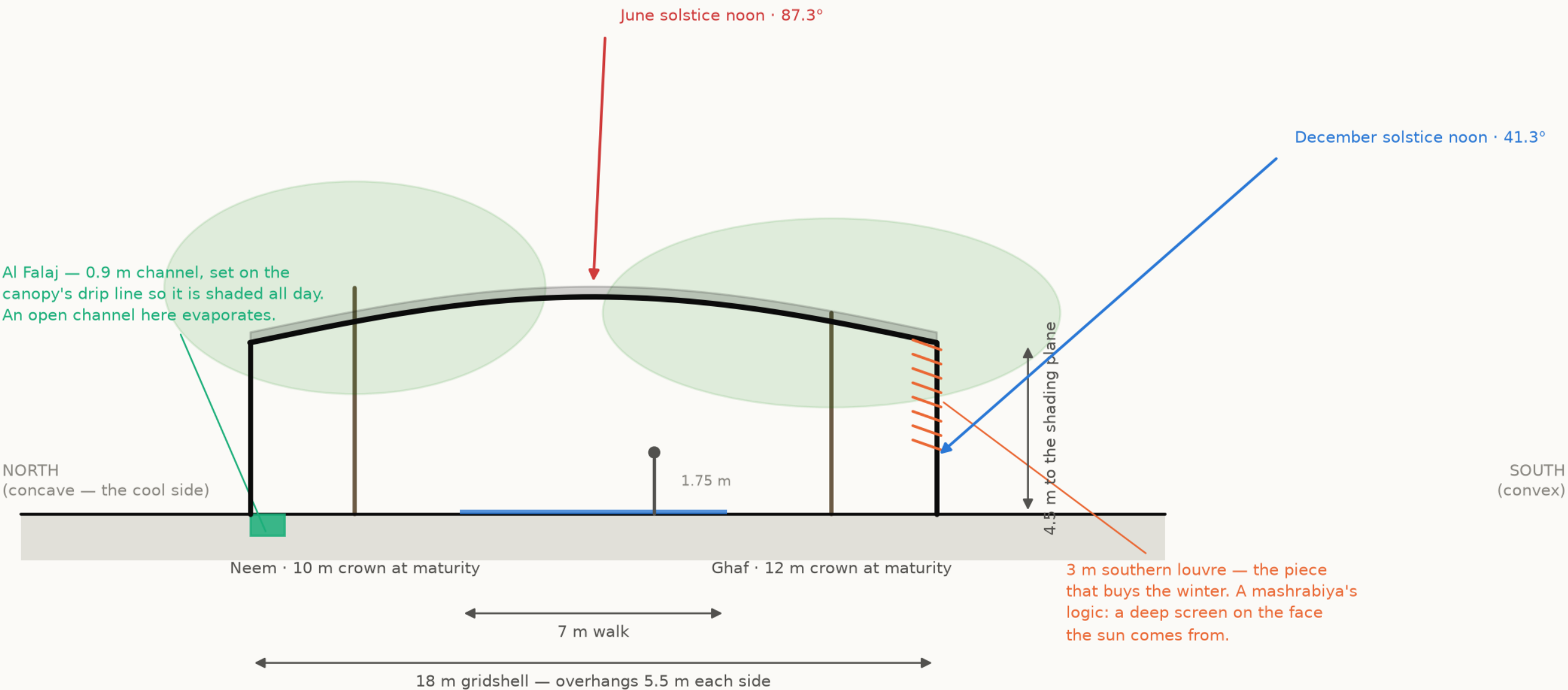
Capital cost plan — 77% of the budget, AED 8.0 M of headroom



Quantities are measured, not estimated: zone areas are the shoelace areas of the drawn polygons, and the canopy is 144 m of true arc x 18 m — not the 138 m chord, which would under-measure it. 12 allowance lines and the three on-cost percentages (preliminaries 12%, contingency 10%, fees 9%) are ASSUMED, and flagged as such in data/processed/cost_plan.csv. AED 1,798/m2 across the 15,000 m2 site. Source: src/costing.py — quantities from the drawn geometry, rates from data/raw/construction_unit_rates_aed.csv · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Section A-A — the Crescent Canopy at midspan

Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N



The plane alone loses the walk to a low southern sun. The louvre is what holds the shadow on the path from November to January.
Source: src/config.py CRESCENT; sun angles NREL SPA via pvlib · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The complete project, and where to check it

Every link below is live. The repository holds the data, the code, the trained models and the tests, and runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA

Live portal wasim437.github.io/AI_SAFA/

START HERE

EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language
PROJECT_PLAN.md	Requirements, phases, status, what is left
README.md	The design argument, written for a juror
notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded

THE GEOMETRY AND THE MODELS

src/plan.py	Single source of the crescent geometry
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/solar.py	Sun position and shadow ray-tracing
src/dataset.py	Assembles the ML training tables
src/models.py	The four models
src/costing.py	The cost model against the AED 35 M ceiling

THE DATA, AS ISSUED AND AS PROCESSED

data/raw/sources.json	Every source dataset, with its period
DATA_SOURCES.md	Sources and their stated limitations
data/raw/site_zoning_schedule.csv	The measured room schedule
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/cost_plan.csv	The capital cost plan, line by line

THE RESULTS, AND THE CHECKS ON THEM

models/model_metrics.json	Trained-model metrics
models/headline_metrics.json	The headline numbers
tests/test_pipeline.py	38 correctness checks
archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why

THE DRAWINGS

figures/fig10_masterplan.png	Masterplan and room schedule
design/visuals/section_crescent.png	Section A–A at midspan
design/visuals/circulation_crescent.png	Circulation and accessibility
design/visuals/facilities_crescent.png	Commercial & Service Facilities Map
design/visuals/planting_crescent.png	Planting plan, 131 trees